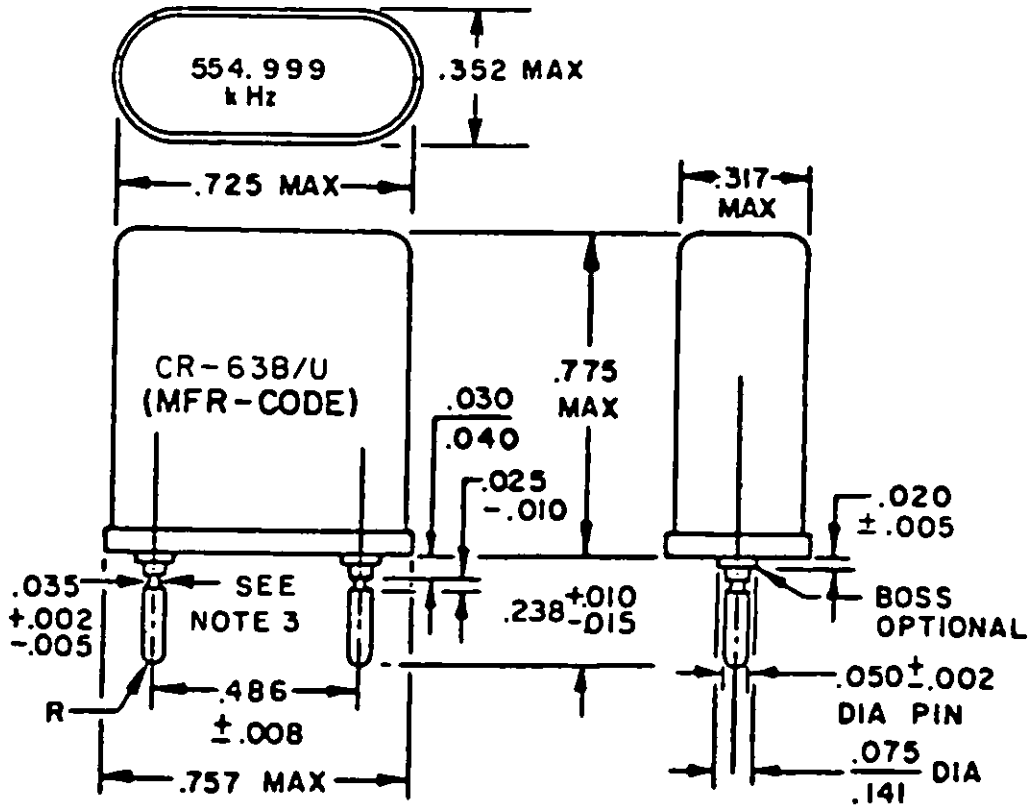
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-63B/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the crystal unit described herein shall consist of this specification and the issue of MIL-C-3098.

Pertinent characteristics: 200 to 555 kHz; fundamental; noncontrolled; antiresonance.



INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.002	.05	.020	.51	.050	1.27	.352	8.94
.005	.13	.025	.64	.075	1.90	.486	12.34
.008	.20	.030	.76	.141	3.58	.725	18.42
.010	.25	.035	.89	.238	6.05	.757	19.23
.015	.38	.040	1.02	.317	8.05	.775	19.68

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. The pin undercut may be omitted.

FIGURE 1. Type CR-63B/U, crystal unit.

Ⓕ denotes changes

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TABLE I. Requirements.

Requirement	Specified value or condition
Holder	HC-6/U (MIL-H-10056/2)
Dimensions, marking, and configuration	See figure
Frequency range	200 to 555 kHz, inclusive
Frequency tolerances:	
Primary operating temperature range	±100 PPM
Secondary operating temperature range	±150 PPM
Equivalent resistance	See table III
Mode of oscillation	Fundamental
Antiresonance, load capacitance	20.0 ± 0.5 pF
Operating temperature ranges:	
Primary	-40° to +70°C, inclusive
Secondary	-55° to -40°C, inclusive, and 70° to 90°C, inclusive
Test set, calibration values, and rated drive level	See table II
Shock (specified pulse):	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	±15 percent
Vibration: 1/	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	±15 percent
(F) Thermal shock:	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	±15 percent
Bond strength:	kHz — Grams, minimum
	200 to 250 — 700
	250+ to 320 — 500
	320+ to 370 — 400
	370+ to 435 — 300
	435+ to 555 — 250

1/ Method 204 of MIL-STD-202, test condition A. 0.060 inch total excursion, 10 to 55 Hz; amplitude to produce 5 gravity units, 55 to 500 Hz.

TABLE II. Test set, calibration values, and rated drive level.

Government reference standard test set	Band of specified frequencies, inclusive	Calibration values			Rated drive level
		Resistance	Crystal current	Resistor voltage drop	
	kHz	Ohms	mA	Volt	mW
TS-710/TSM	200 to 225	3700	---	2.72	2.0 ±0.4
	225+ to 275	4200	---	2.90	
	275+ to 325	4600	---	3.03	
	325+ to 375	4900	---	3.13	
	375+ to 425	5300	---	3.26	
	425+ to 475	5600	---	3.35	
	475+ to 500	6000	---	3.46	
	500+ to 555	3500	---	2.65	

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TABLE III. Equivalent resistance.

Frequency range, inclusive	Maximum resistance
kHz	Ohms
200 to 225	5, 300
225+ to 275	6, 000
275+ to 325	6, 500
325+ to 375	7, 000
375+ to 425	7, 500
425+ to 475	8, 000
475+ to 500	8, 500
500+ to 555	5, 000

NOTES:
1. The socket designed to mate with crystal holder HC-6/U is Part No. M12883/04-01 (see MIL-S-12883/4).
   For Navy use, other sockets listed in MIL-STD-242 may be used.
2. NATO preferred type NXT23/A.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI, AR, SG, SM
Air Force - 11, 17, 99
DLA - ES

User activities:
Navy - AS, MC, SH
Air Force - 17, 19

Agent:
DLA - ES

Preparing activity:
Army - ER

(Project 5955-0606-2)

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